Hello

Does longer context help a small character-level transformer?

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Research question

Does seeing more previous characters improve next-character prediction?

32 128 256 characters

Keep the number of training characters fixed.

Longer context costs more compute. Does it earn a better score?

Model and controlled comparison

GPT-style decoder: 4 layers, 4 heads, width 192
About 1.8M parameters. AdamW. 3,000 steps.

Context length	Windows per batch	Characters per step
32	512	16,384
128	128	16,384
256	64	16,384

References: context 32 and a tuned character n-gram

Data and evaluation

TinyShakespeare: 1.1M characters, vocabulary of 65

81% train / 9% validation / 10% fixed test
10 validation folds, matched seeds, 30 transformer runs

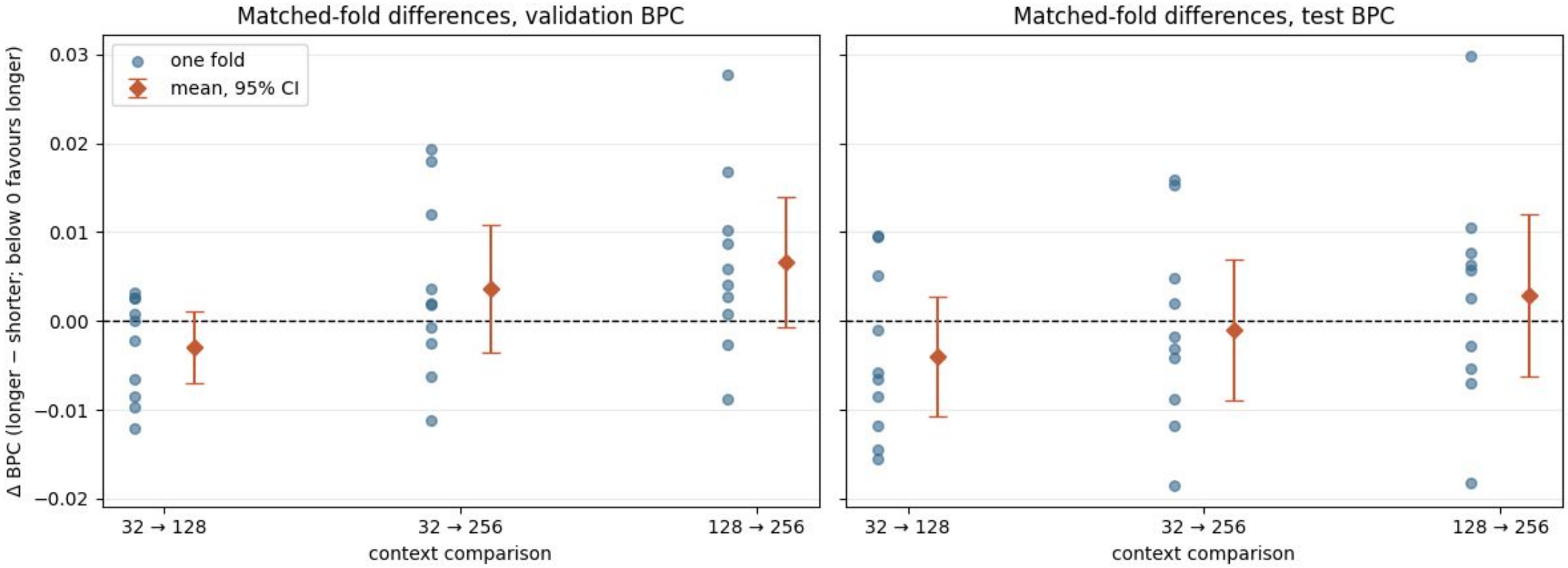
Bits per character (BPC): lower is better

Score the same targets with a full context window.
Compare longer and shorter models within each fold.

No convincing overall improvement

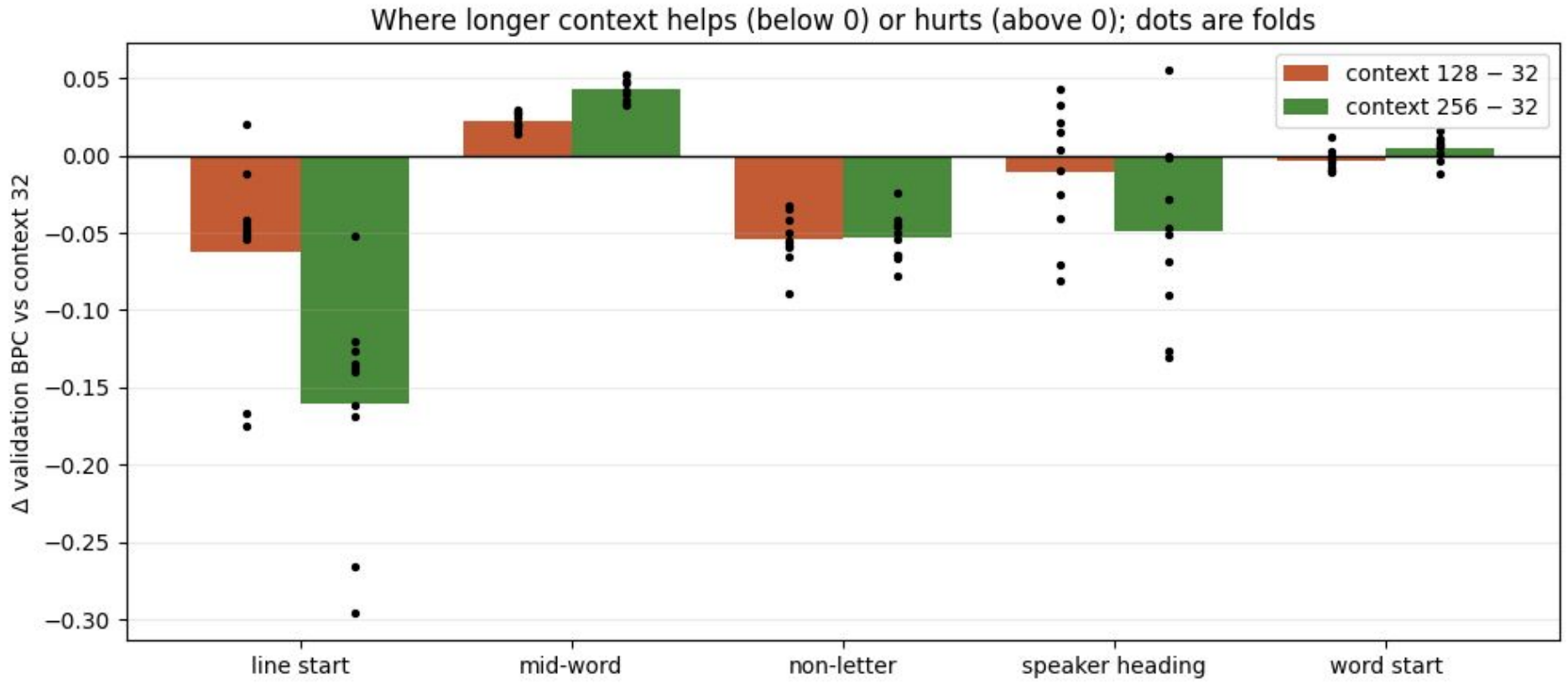
Mean test BPC: 32 = 2.1335 128 = 2.1295 256 = 2.1324

Tuned n-gram = 2.4954. All paired 95% intervals include zero.



Longer context helps some patterns

Validation differences relative to context 32. Below zero is better.



Mid-word letters are 51.5% of targets, so their losses offset the gains.

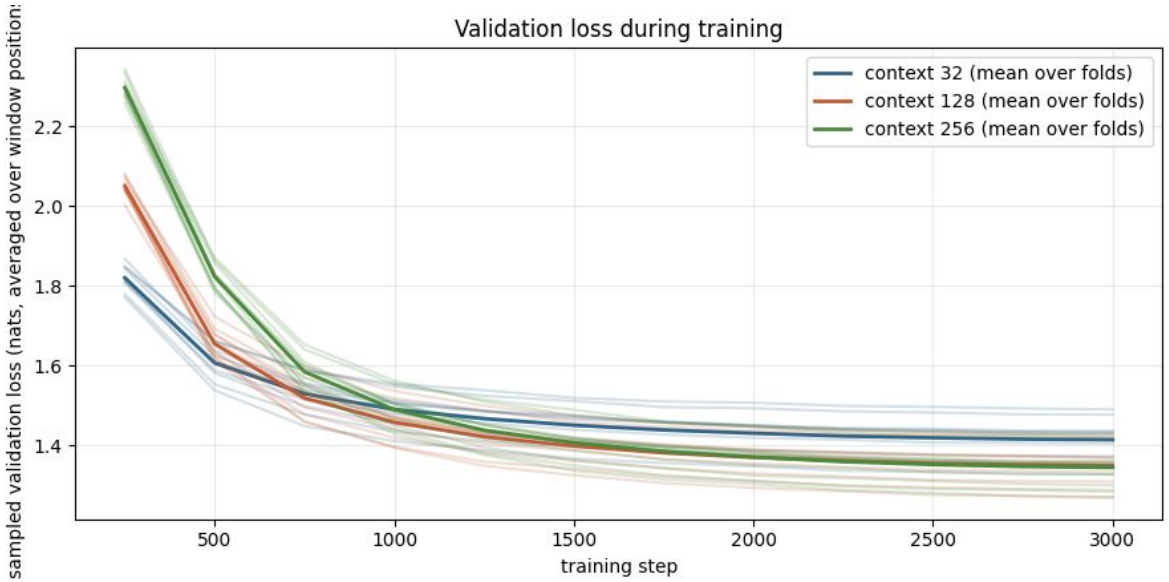
Conclusion and limitations

No convincing overall gain from longer context at this model size and training budget.

27 of 30 runs chose the final checkpoint.

Fold 9 shows sensitivity to familiarity with the test play.

Next: a larger model and longer training.



Window-average loss can favour longer contexts.

Thank you

Questions?

Daniel Moody and Nandakishore Vinayakrishnan